

Trauma General Patient Care Guideline

Key Concepts

- **Penetrating trauma** – perform MARCHES and begin evacuation to closest capable facility (preferred Level 1 or 2 trauma center)
- **Blunt traumatic arrest** – perform MARCHES and if no response consider termination of resuscitation protocol
- Backboards with full c-spine immobilization IS NOT RECOMMENDED in penetrating trauma

MARCHES Protocol

- **Massive Bleeding control**
- **Airway** – NPA/OPA/Cricothyrotomy
- **Respiratory** – decompress chest if tension pneumothorax suspected, occlusive dressing for open pneumothoraces
- **Circulation**- IV/IO, pelvic binder, Reassess tourniquet/wound packing
- **Hypothermia care/Hike** vs helicopter (consider evac options)
- **Eye injuries** – cover with rigid shield and no pressure on the eye
- **Spinal motion restriction** if indicated

EMT/AEMT

- Stop **Massive Bleeding** as first action
 - Tourniquet
 - Compress large vessel in the wound by hand against bone
 - Wound packing
 - Apply pressure dressing with firm direct pressure
- Establish Simple **Airway** with NPA or OPA and suction secretions if needed
- Support **Respirations**
 - BVM with **Oxygen**, if needed
- Support **Circulation**
 - Reassess tourniquets, bleeding wounds
 - Use pelvic binder for pelvis fractures
 - If in arrest, start CPR, may establish IO access
- **Hypothermia**- after primary and secondary head to toe assessment is completed- remove wet clothing and cover with blankets.
- Penetrating **Eye Injuries** should be covered with eye shield with zero pressure on the eye
- **Spinal Motion Restriction Guideline** should be considered for falls, MVC and blunt trauma
 - **Continuously monitor tourniquet and wound packing interventions****
 - Reassess after every patient move**

EMT/AEMT - Expanded Scope

- **Igel** if needed for ventilation support
- **NS/LR** 500ml IO if BP $\leq$ 90 Systolic
- **TXA** per TXA Guideline
- Pain per Pain Management Guideline - Avoid Ibuprofen and Ketorolac in patients with potential for bleeding

Paramedic

- **Airway** – consider advanced airway management or surgical cricothyrotomy
 - Capnography is required if available
- **Respiration** – consider needle thoracostomy if unable to ventilate effectively or in profound shock/arrest with thoracic trauma
 - 4-5th intercostal space anterior axillary line (nipple level) above rib is preferred location
 - 2nd intercostal space, mid-clavicular is alternative location
- **Circulation** – IV/IO with boluses of fluid, **Lactated Ringers (LR)** for burn patients and **Normal Saline (NS)** if LR not available
 - **TXA** per TXA Guideline
 - Pelvic binder for pelvic instability and reassess all bleeding wounds for hemostasis

Trauma Fluid Administration Guidelines—Administer appropriate fluid if any of the following are met

- Systolic Blood Pressure < 70 mmHg **OR**
- Systolic Blood Pressure < 90 mmHg with Heart Rate ≥ 110 beats per min **OR**
- ETCO2 < 25 **OR**
- Witnessed traumatic arrest < 5 min prior to provider arrival and continuous CPR throughout downtime **OR**
- Age ≥ 65 yo and SBP ≤ 100 **AND** HR ≥ 100 beats per minute
- **Pain** – consider Pain Management Guideline - Avoid Ibuprofen and Ketorolac in patients with potential for bleeding
- **Spine** – maintain spinal motion restriction, if indicated